

Relational Model and Database Design

Unit II

Relational Database Schema

- **Definition:**

A relational database schema is a **logical structure** of a database that defines **tables, attributes, relationships, and constraints**.

- **Components of Relational Schema**

- **Relation (Table Name)**

- **Attributes (Columns)**

- **Domains (Data Types)**

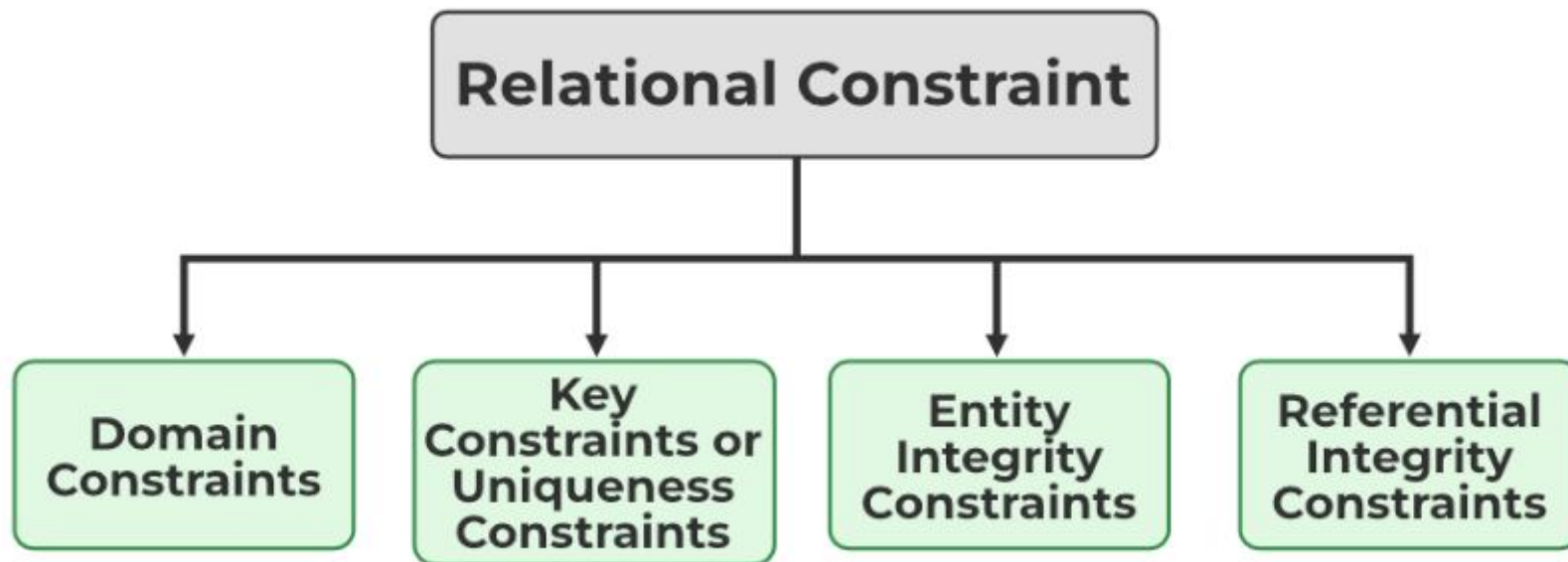
- **Primary Key**

- **Foreign Key**

- **Constraints**

- In modeling the design of the relational database we can put some restrictions like what values are allowed to be inserted in the relation, and what kind of modifications and deletions are allowed in the relation.
- These are the restrictions we impose on the relational database.

- **Superkey** – Any set of attributes that uniquely identifies a tuple in a relation.
- **Candidate key** – A minimal superkey. Removing any attribute would break the uniqueness property.
- **Primary key** – A chosen candidate key that uniquely identifies each tuple in the table.



1. Domain Constraints

- Every domain must contain atomic values (smallest indivisible units) which means composite and multi-valued attributes are not allowed.
- We perform a datatype check here, which means when we assign a data type to a column we limit the values that it can contain.
- Eg. If we assign the datatype of attribute age as int, we can't give it values other than int datatype.

Example:

EID	Name	Phone
01	Bikash Dutta	123456789 234456678

Key Constraints or Uniqueness Constraints

- These are called uniqueness constraints since it ensures that every tuple in the relation should be unique.
- A relation can have multiple keys or candidate keys(minimal superkey), out of which we choose one of the keys as the primary key, we don't have any restriction on choosing the primary key out of candidate keys
- Null values are not allowed in the
- primary key,
- Hence Not Null constraint is also part of the key constraint.

Example:

EID	Name	Phone
01	Bikash	6000000009
02	Paul	9000090009
01	Tuhin	9234567892

Entity Integrity Constraints

- Entity Integrity constraints say that no primary key can take a NULL value, since using the primary key we identify each tuple uniquely in a relation.
- In the above relation, EID is made the primary key, and the primary key can't take NULL values but in the third tuple, the primary key is null, so it is violating Entity Integrity constraints.

Example:

EID	Name	Phone
01	Bikash	9000900099
02	Paul	6000000009
NULL	Sony	9234567892

Referential Integrity Constraints

- The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.
- This constraint is enforced through a foreign key, when an attribute in the foreign key of relation R1 has the same domain(s) as the primary key of relation R2, then the foreign key of R1 is said to reference or refer to the primary key of relation R2.
- The values of the foreign key in a tuple of relation R1 can either take the values of the primary key for some tuple in relation R2, or can take NULL values, but can't be empty.

Example

- DNO of Table 1 is the foreign key, and DNO in Table 2 is the primary key. DNO = 22 in the foreign key of Table 1 is not allowed because DNO = 22 is not defined in the primary key of table 2. Therefore, Referential integrity constraints are violated here.

Example:

EID	Name	DNO
01	Divine	12
02	Dino	22
04	Vivian	14

DNO	Place
12	Jaipur
13	Mumbai
14	Delhi

Advantages of Relational Database Model

- **Data Integrity:**
- **Flexibility:**
- **Scalability:**
- **Security:**
- **Data consistency:**
- **Query Optimization:**

Disadvantages of the Relational Model

- **Complexity:**
- **Performance:**
- **Scalability:**
- **Cost:**
- **Limited flexibility:**
- **Data redundancy**

